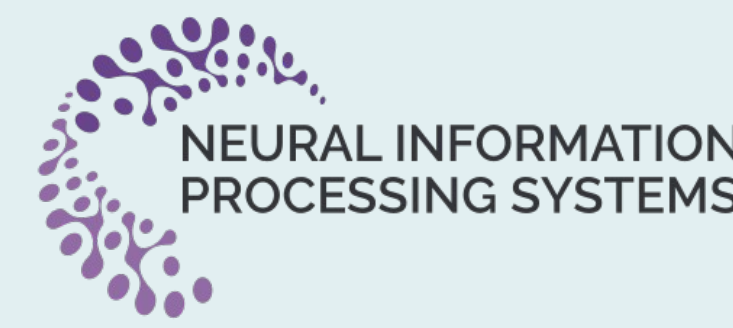


Flexible Attention-Based Multi-Policy Fusion for Efficient Deep Reinforcement Learning

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Problem

RL agents are **sample-inefficient**; injected *knowledge policies* stay rigid and can't be recombined or replaced across tasks.

HUMANS	VS	RL AGENTS
reuse & recombine freely		fixed order → retrain to edit

Motivation

- Treat external policies as a **modular, reorderable set**.
- Prior fusion (KoGuN, A2T) fixes policy order, forcing **retraining**.

sample-efficient generalizable

compositional incremental

Contribution

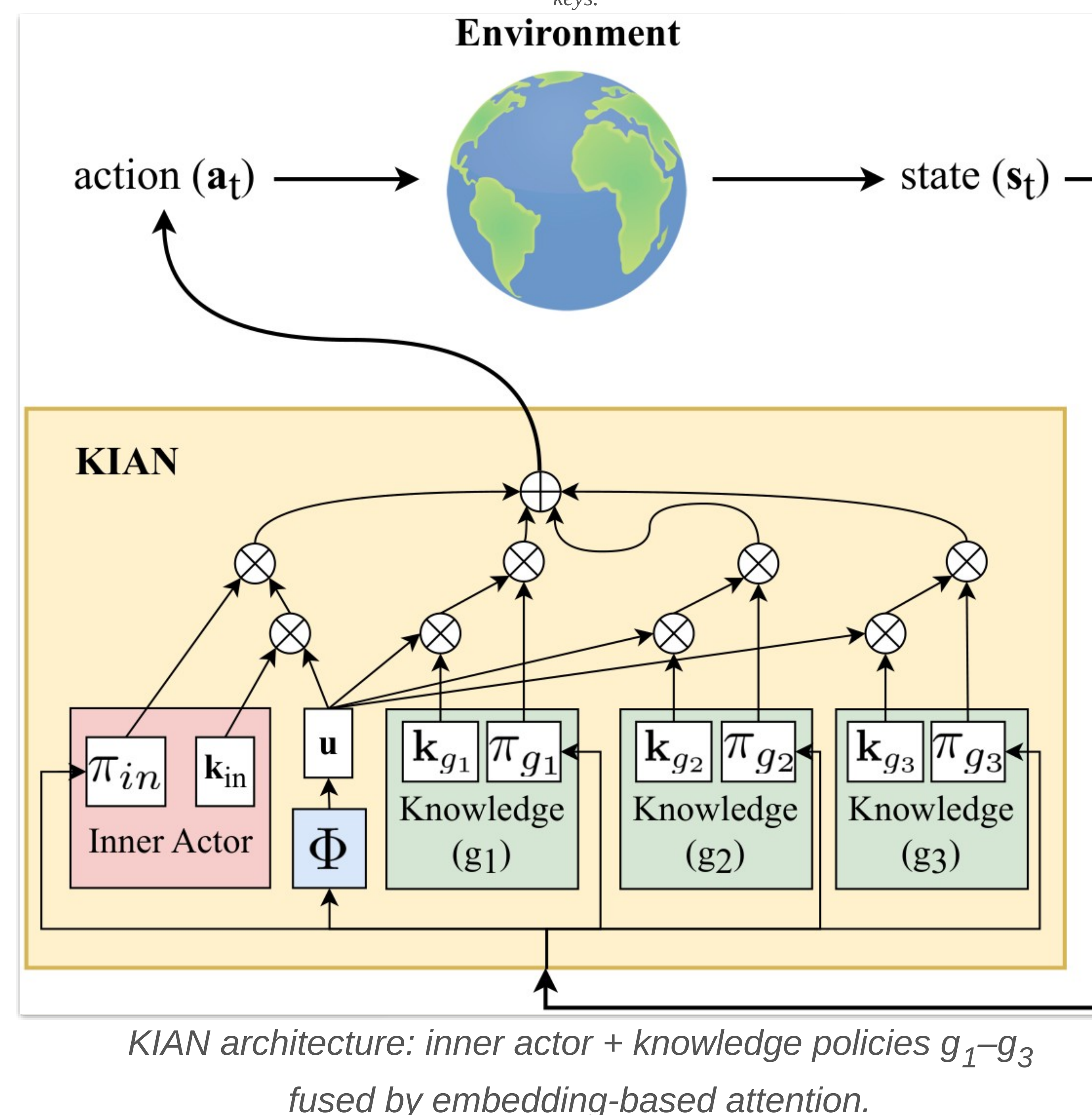
- KGRL** — fuse an inner policy with many knowledge policies.
- KIAN** — attention rearranges, adds, or replaces policies.
- Entropy imbalance** — identified, proven, and fixed.

Method

- KIAN** = an inner actor (the agent's own policy) + a learnable *key* per knowledge policy + a state-dependent *query*.
- The **fused policy** is a weighted sum; keys are unordered, so swap a policy by swapping its key — no retraining.

$$\pi(\cdot | s_t) = \hat{w}_{in} \pi_{in}(\cdot | s_t) + \sum_{j=1}^n \hat{w}_{g_j} \pi_{g_j}(\cdot | s_t)$$

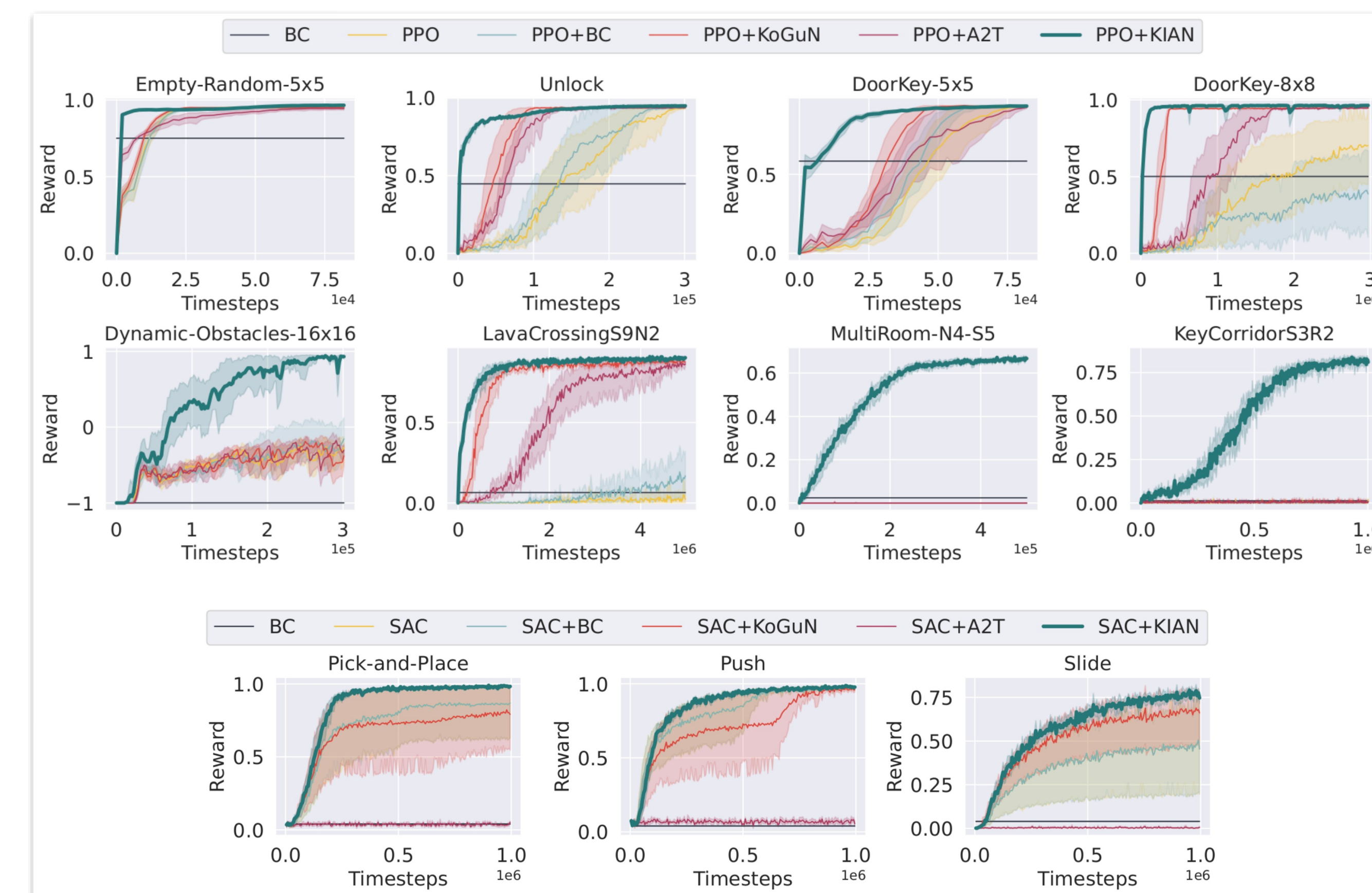
$\hat{w}_{g_j} = \text{softmax}(\Phi(s_t) \cdot k_j / c)$ over the inner and knowledge keys.



Key Results

Zero-shot transfer	Best baseline	KIAN
Empty-Random 5x5 → 16x16	0.53	0.93
Pick-and-Place, 10x goal range	0.30	0.72
DoorKey-8x8 → Reach (C2S)	0.80	1.00

Only method to succeed in **every** environment given a sub-optimal knowledge set — the advantage grows as tasks get harder.



Sample-efficiency learning curves across MiniGrid (top two rows) and OpenAI-Robotics (bottom). KIAN (teal) reaches high reward with far fewer samples on every task.

SO WHAT → Best zero-shot transfer in both simple → complex and complex → simple settings, with lower variance than every baseline (BC, RL, RL+BC, KoGuN, A2T). Methods that ignore entropy imbalance (A2T) barely learn in continuous control, while KIAN keeps improving.

Headline Numbers

0.93 Zero-shot reward, 5x5 → 16x16 (vs 0.53 baseline)	0.72 Pick-and-Place 10x (vs 0.30)
	1.00 DoorKey → Reach (vs 0.80)
	10 seeds · 95% CI

Ablation

- Removing the **entropy-imbalance fix** collapses KIAN onto one policy.
- Compositional & incremental**: reused keys learn tasks sequentially with fewer samples.

SO WHAT → Modular knowledge keys pay off across transfer settings.

Takeaway

Encoding each policy as an independent learnable **key** and fusing them by attention makes external knowledge fully *modular*; the entropy fix keeps exploration efficient. Knowledge policies become **plug-and-play** building blocks.

Add, drop, or reorder knowledge policies anytime — without retraining the network.